

Mini-Challenge 2

Kickoff: Images

**gbsv: Foundation in Image and
Signal Processing
HS25**

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21. April 2026



Mini-Challenge 2 (MC2): Image Processing Methods

**MC1**

- 1D
- Signals

**MC2**

- 2D
- Images

MC2 Topics: Images

Image Augmentation

Pattern Detection:
Edges, Lines, Circles...

Classic Segmentation
Morph. Operators

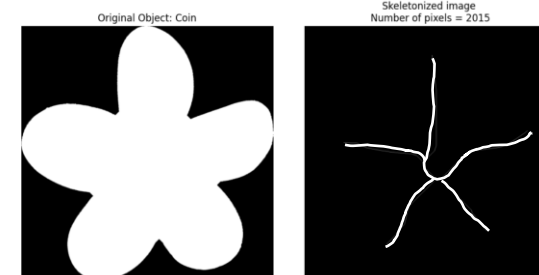
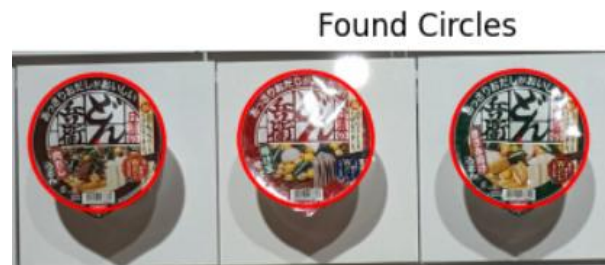
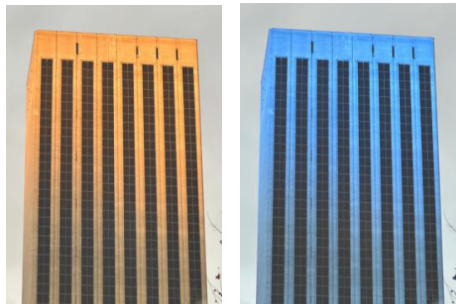
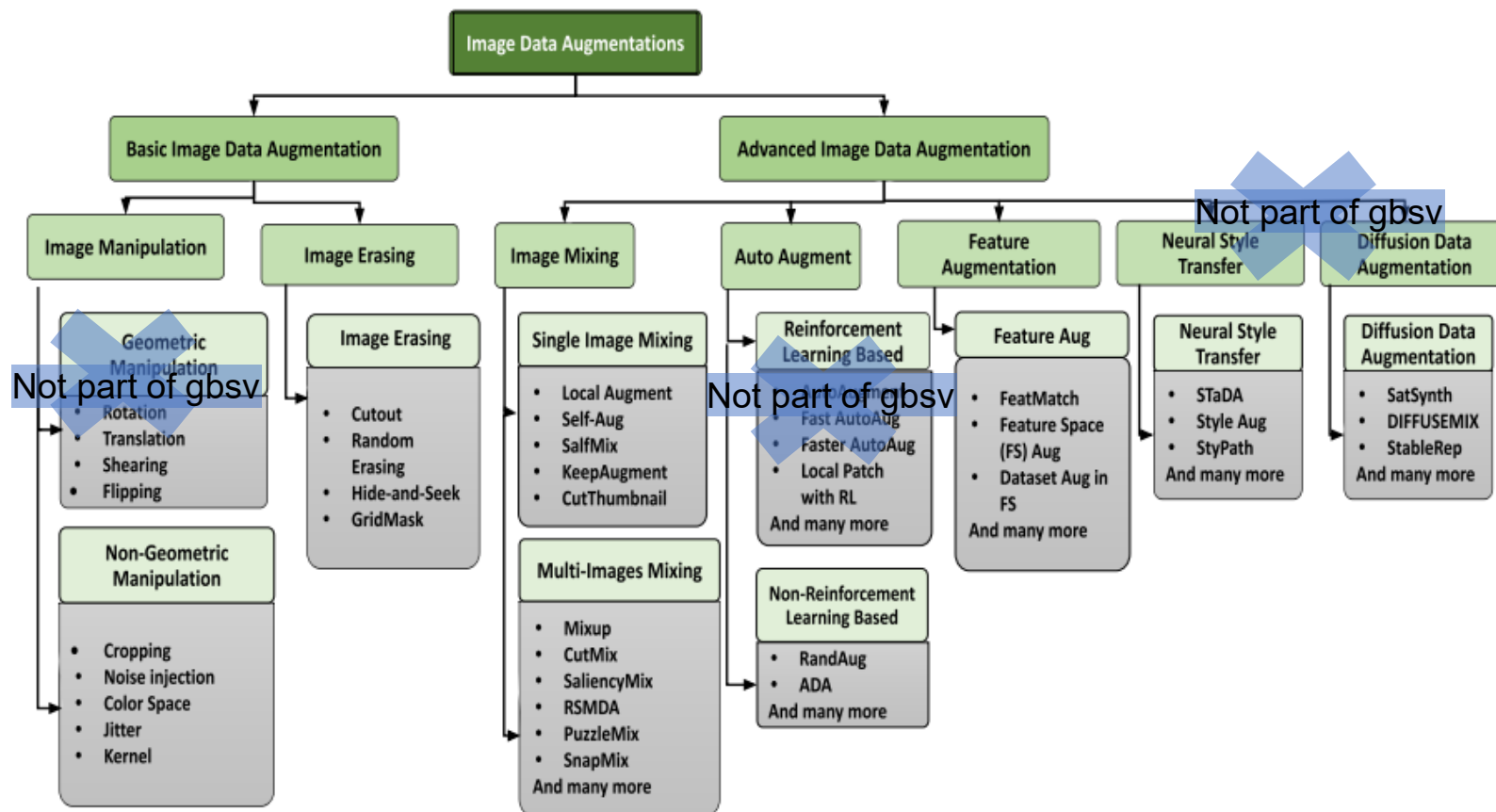


Image Augmentation

Data Augmentation

- NO Deep Learning
- NO geometric manipulations



Sharpening as Augmentation

Originalbild
RMSE: 0.00, PSNR: inf



Sharpness
RMSE: 9.76, PSNR: 12.92



Sharpening as Augmentation

Original (4x Zoom)



Geschärft (4x Zoom)



Noise Injection as Augmentation

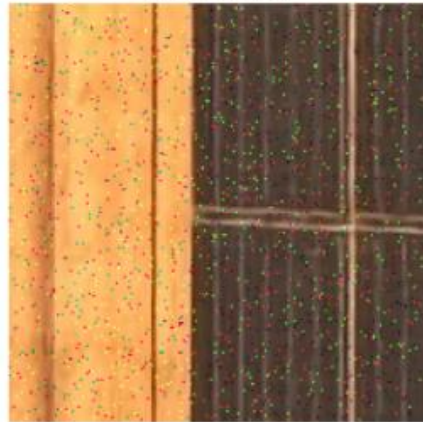
Original Crop



gaussian



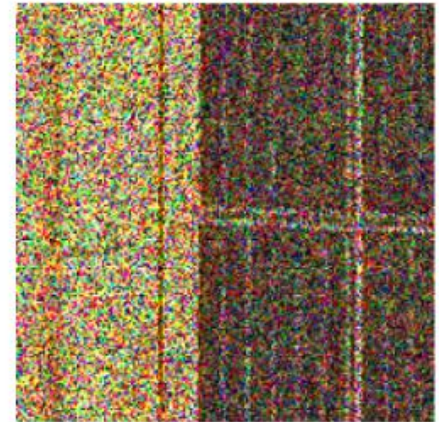
salt_pepper



poisson



speckle



Denoising as Augmentation

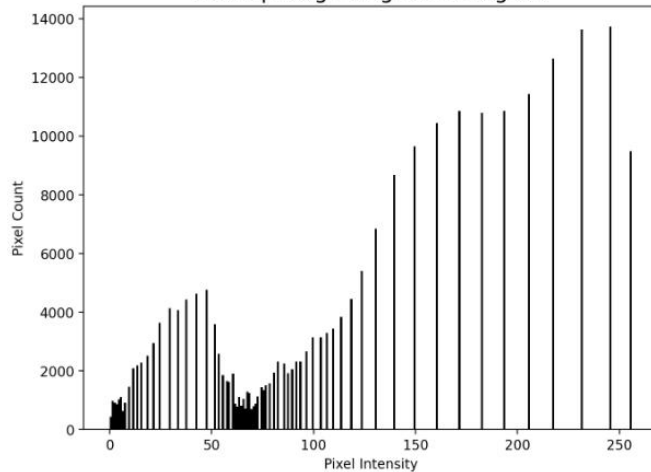
Non-Local Means Denoising
PSNR: 36.1509
SSIM: 0.8440

Bilateral Filtering
PSNR: 36.4481
SSIM: 0.8741

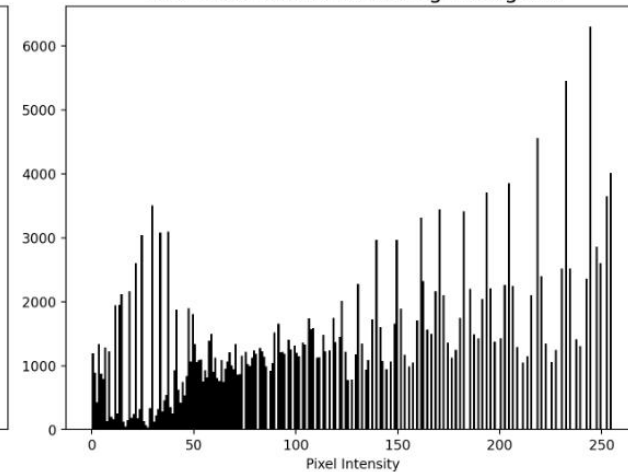
Hist Eq Image Original



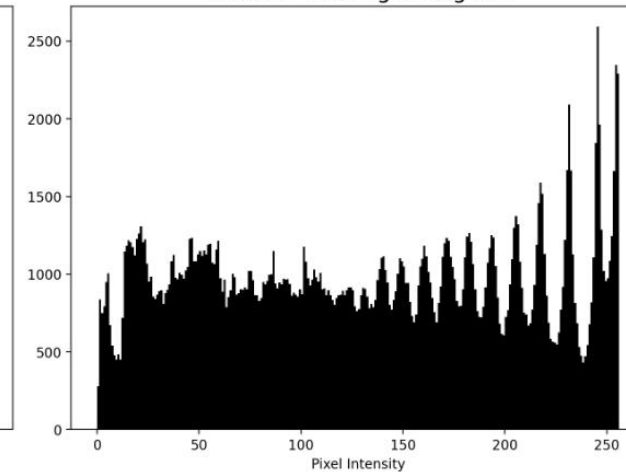
Hist Eq Image Original Histogram



Non-Local Means Denoising Histogram



Bilateral Filtering Histogram



Histogram-Equalization as Augmentation

Namsan Tower Original



Histogram Equalization
PSNR: 12.7939
SSIM: 0.6694



CLAHE
PSNR: 27.7331
SSIM: 0.9626



Lab Color Space
PSNR: 12.6517
SSIM: 0.6431

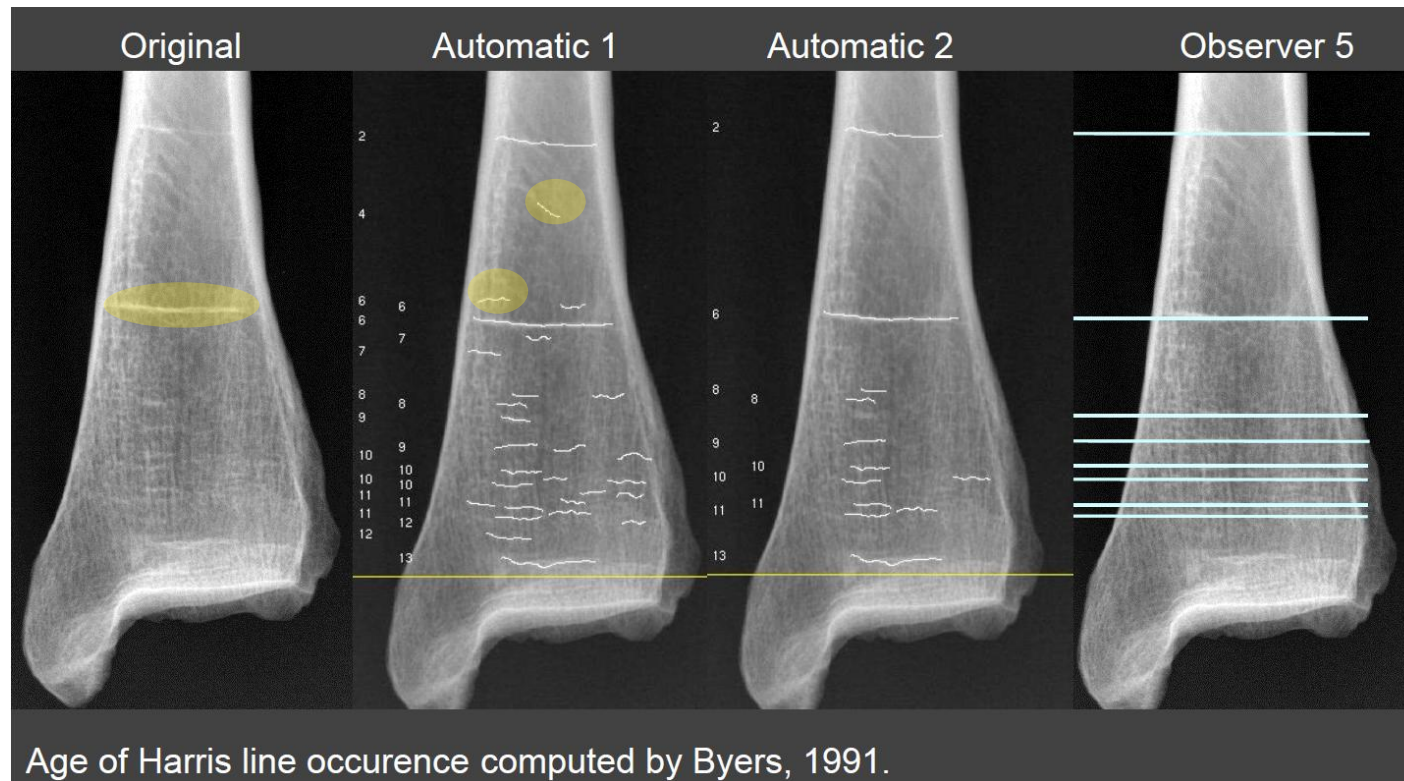


Links

- [See Video Inputs](#)
- [Ultralytics Augmentation](#)

Pattern Detection

Case Study Line Detection: Stress Markers in X-Ray Images of Archaeological Samples

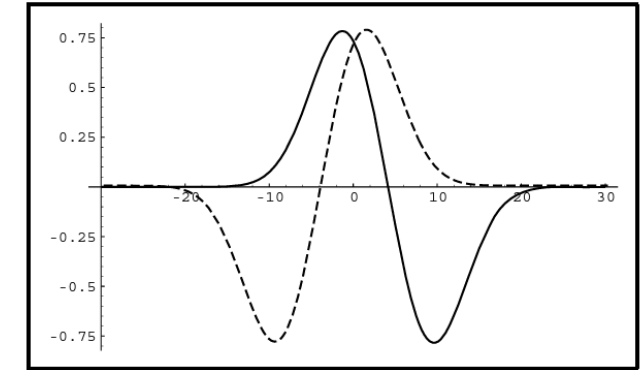


Suter et al. 2008: Technical note: standardized and semiautomated Harris lines detection, https://github.com/susuter/harris_lines_tool, www.harrislinestool.com

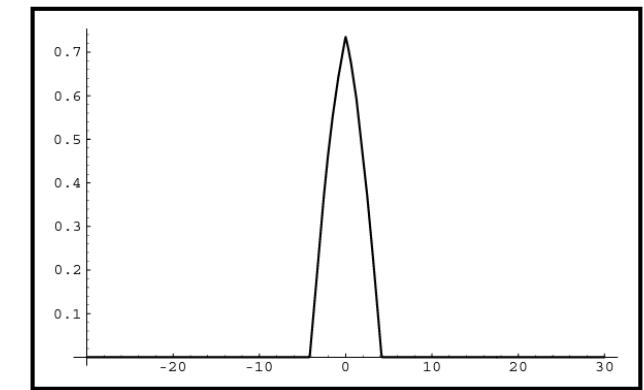
Line Detection - Classic

- Line detection algorithm:
- A line shows two parallel intensity changes / edges
- Line itself is located between these edges
- Find horizontal lines based on 1st derivative and Gaussian
- Noise reduction [Perona & Malik, 1990]

Left and right edges



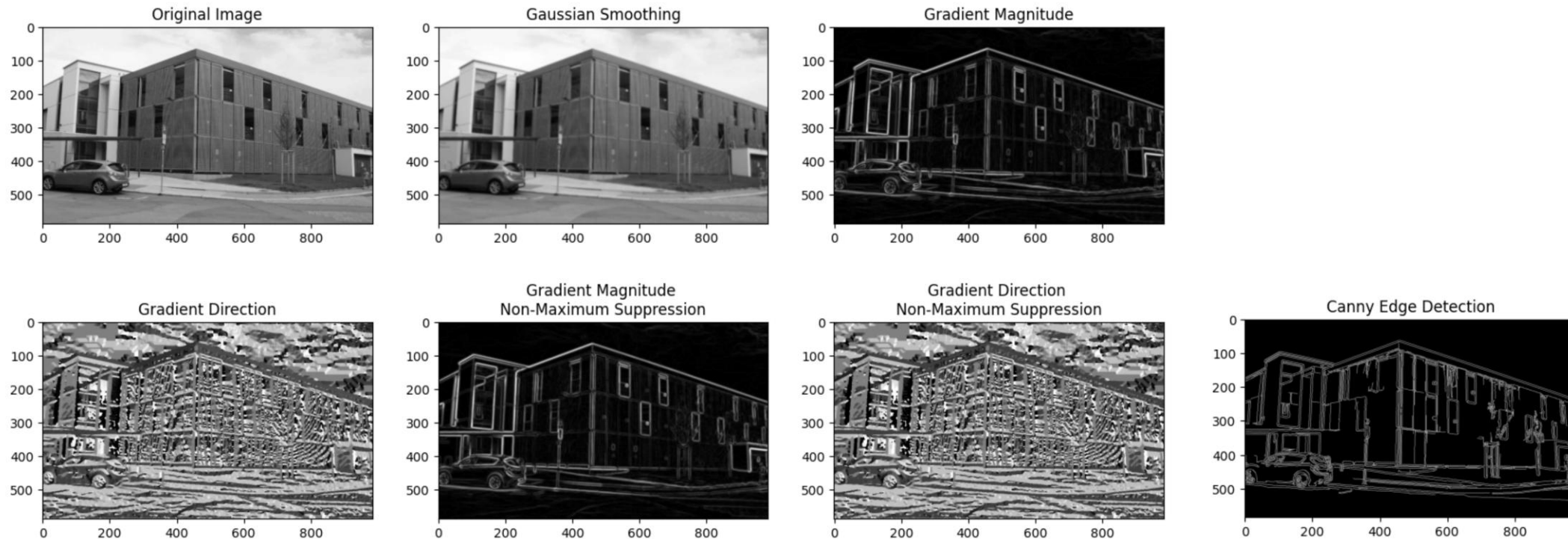
Non-linear combination:
Peak between edges



Koller et al., 1995: Multiscale detection of curvilinear structures in 2D and 3D image data

Suter et al. 2008: Technical note: standardized and semiautomated Harris lines detection, https://github.com/susuter/harris_lines_tool

Canny Edge Detection



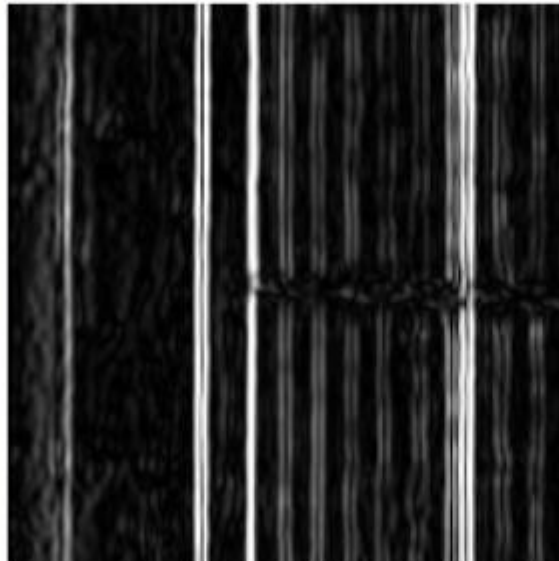
Hinweis: Hysterese fehlt

Line Detection

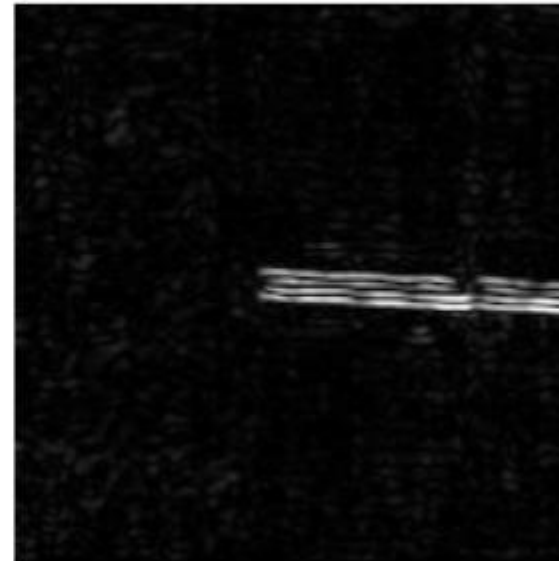
Cropped (RGB)



Sobel X



Sobel Y



Gradient Magnitude



Edge / Line Detection

- [Lesespur: Ecken- und Kantenfilter \(Deutsch\)](#)
- <https://blog.roboflow.com/edge-detection/>

Segmentation & Morphological Operators

see separate input & hands-on notebooks

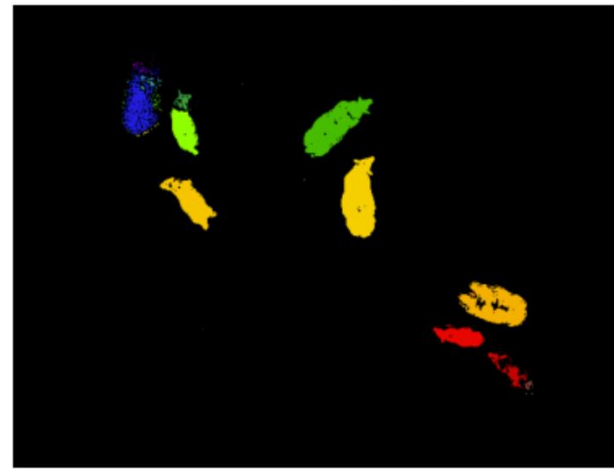




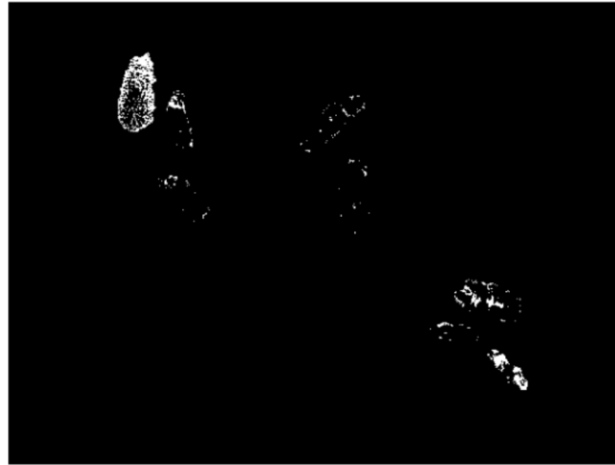
Originalmaske HSV



Nach Closing



Unterschied (Closing)

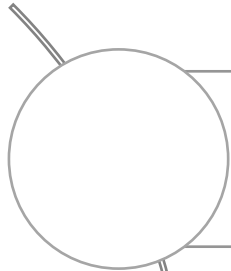


Skeletons aller Objekte mit Objekt-Nummern

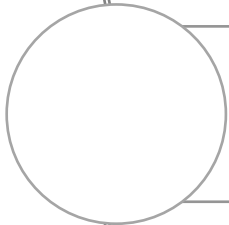


Guidelines / FAQs

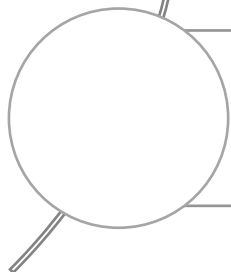
What is important during gbsv MCs?



Use-case driven design: Data, methods, and experiments are well aligned with the problem



Coherent experimental setup: Small number of clearly structured, tightly coupled experiments (no loosely connected variants)



Analytical and critical reasoning: Results are thoughtfully analyzed with targeted creativity and clear, concise communication

Appropriate Region of Interest Selection



Original Crop



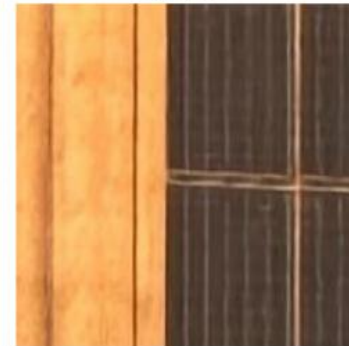
gaussian



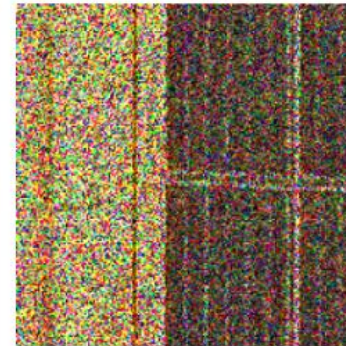
salt_pepper



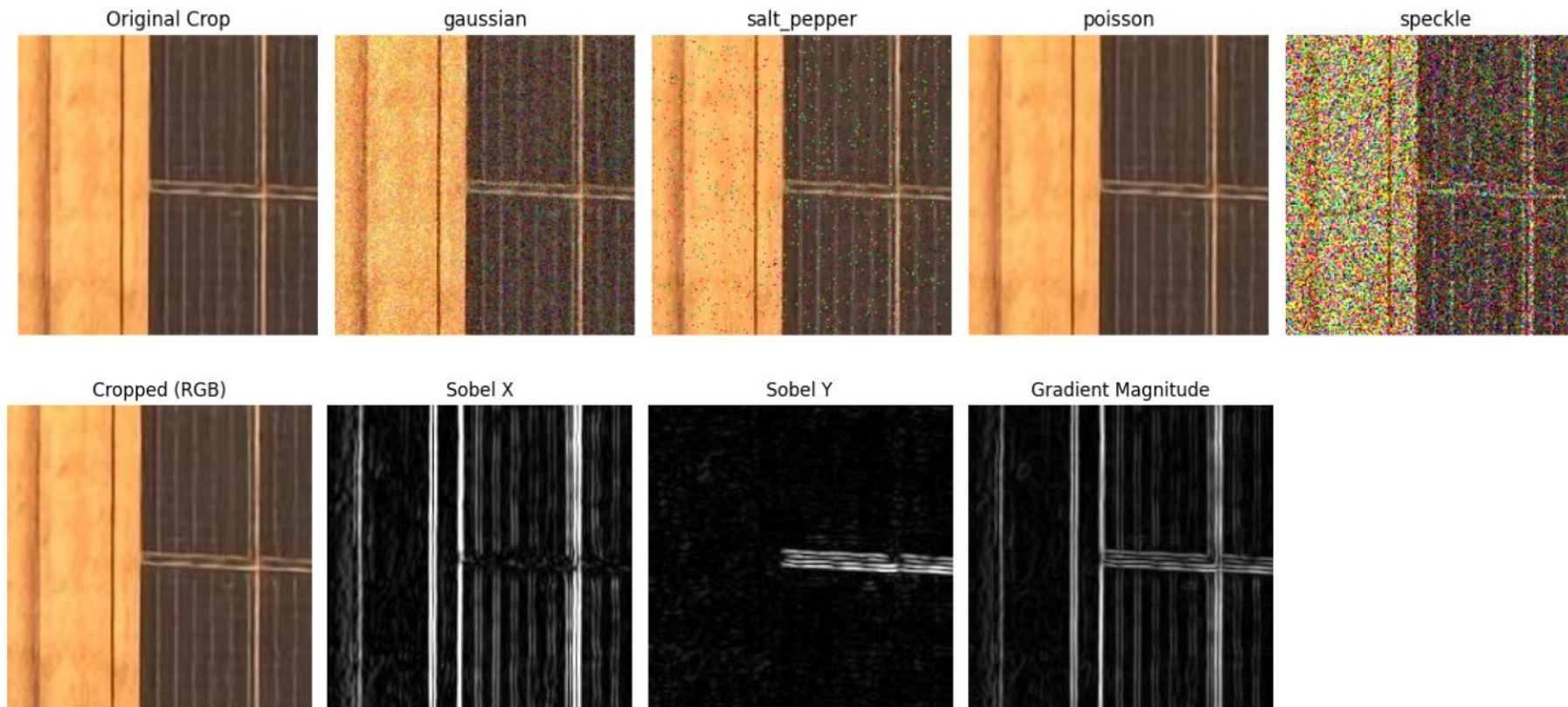
poisson



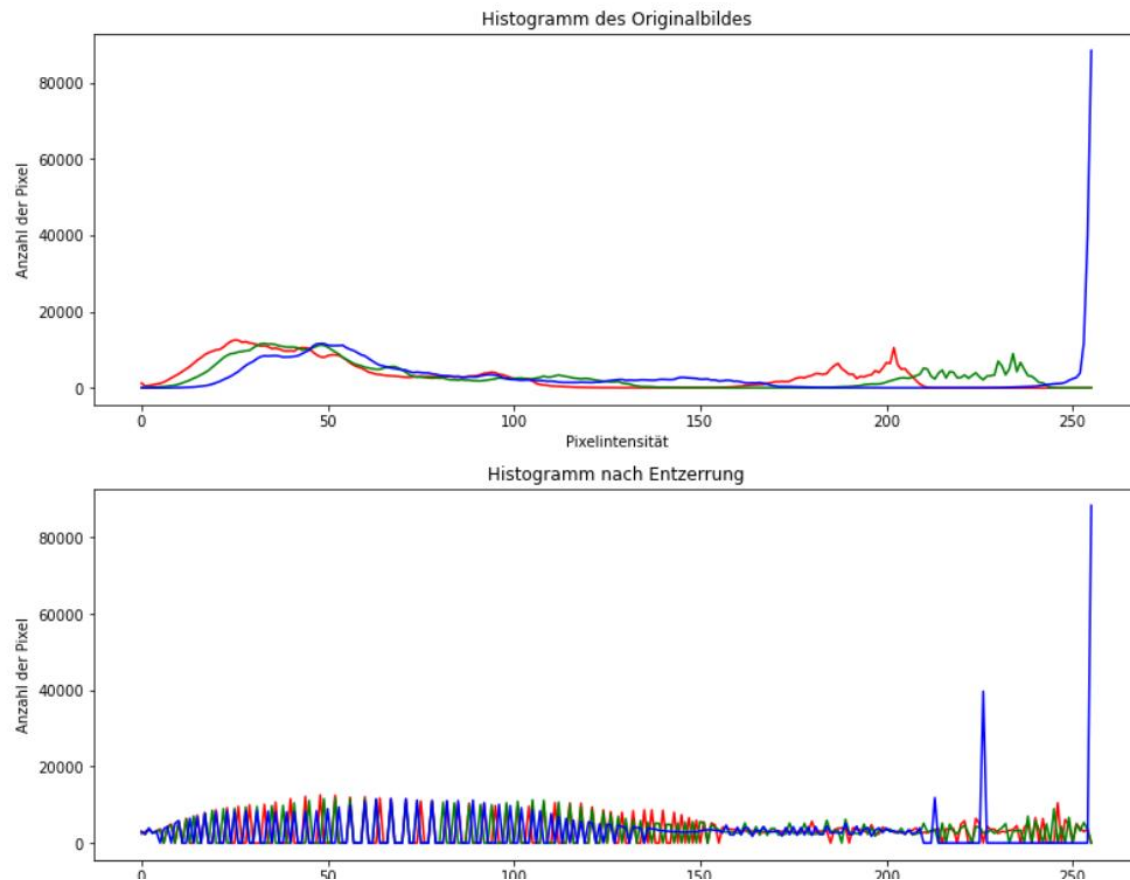
speckle



Data May Be Reused for Varying Sub Tasks



Accompanying KPIs with Figures



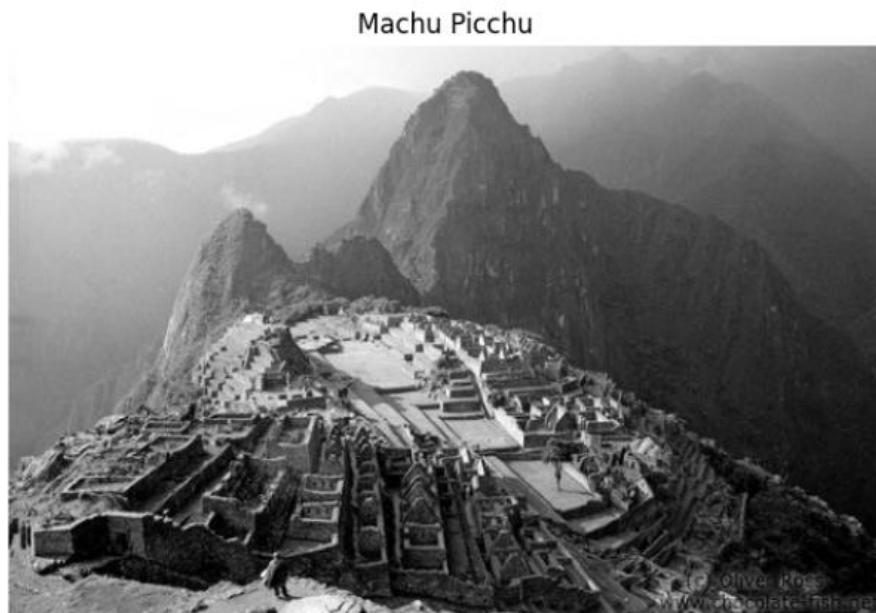
Originalbild - Wasserfall



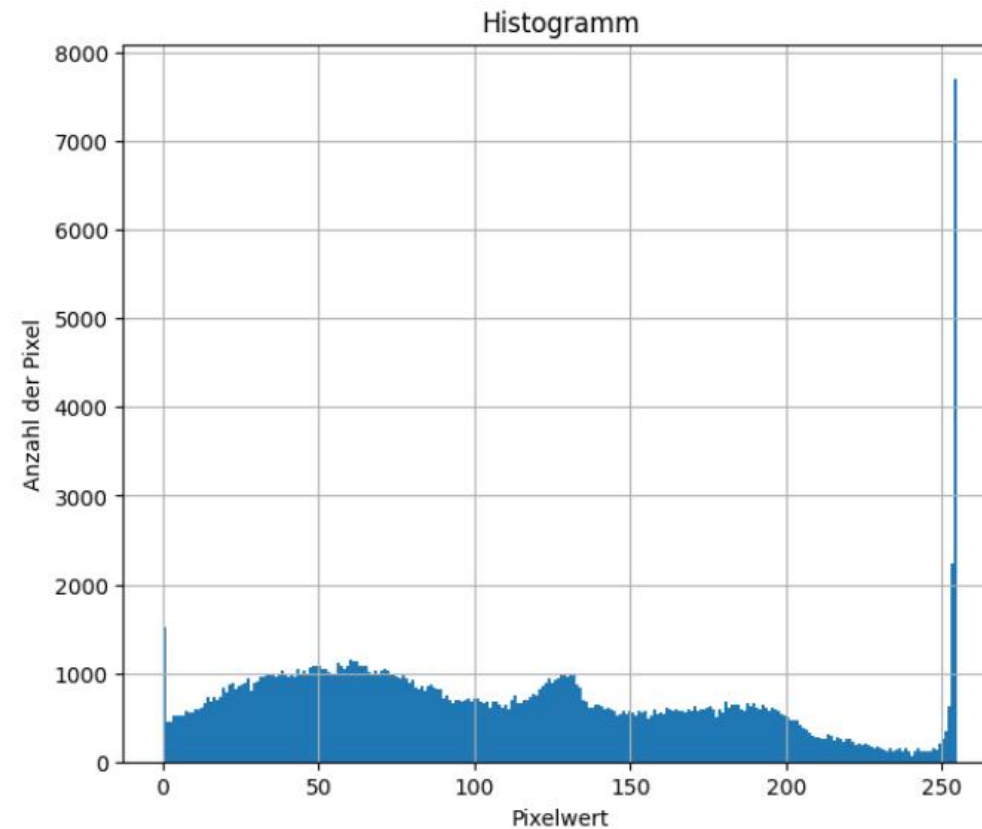
Bild nach Histogramm-Entzerrung



Accompanying KPIs with Figures

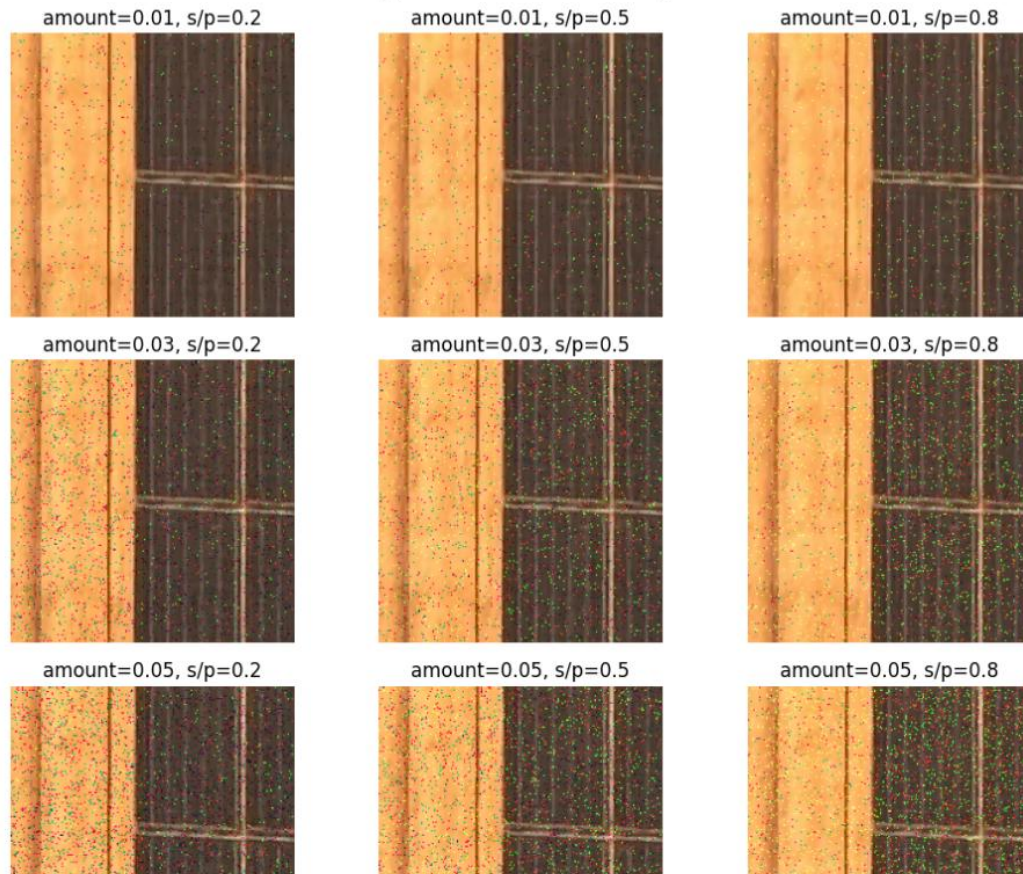


Helligkeit: 110.64
Schärfe: 90.03
Wertebereich: (0, 255)



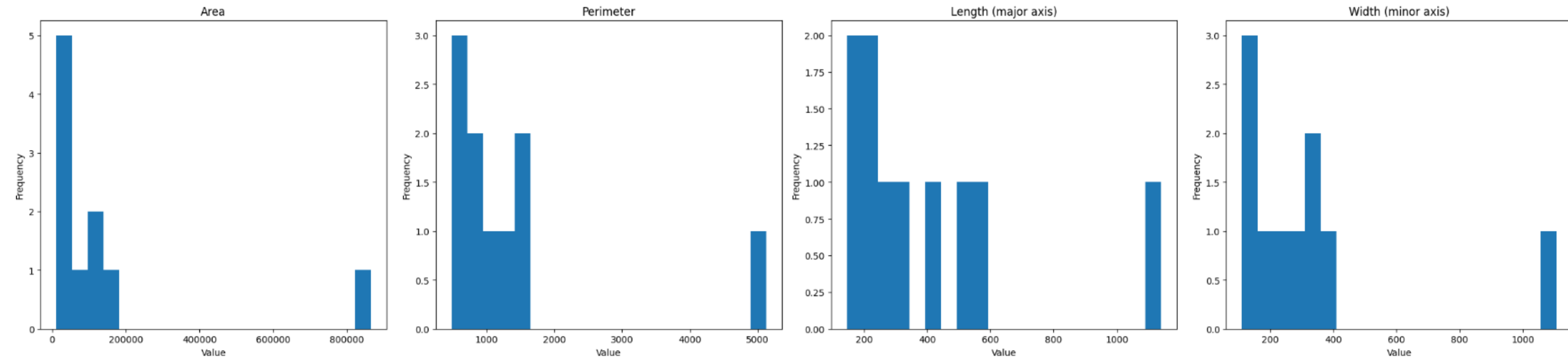
Explore Parameter Spaces

Salt & Pepper Noise Parameter Exploration



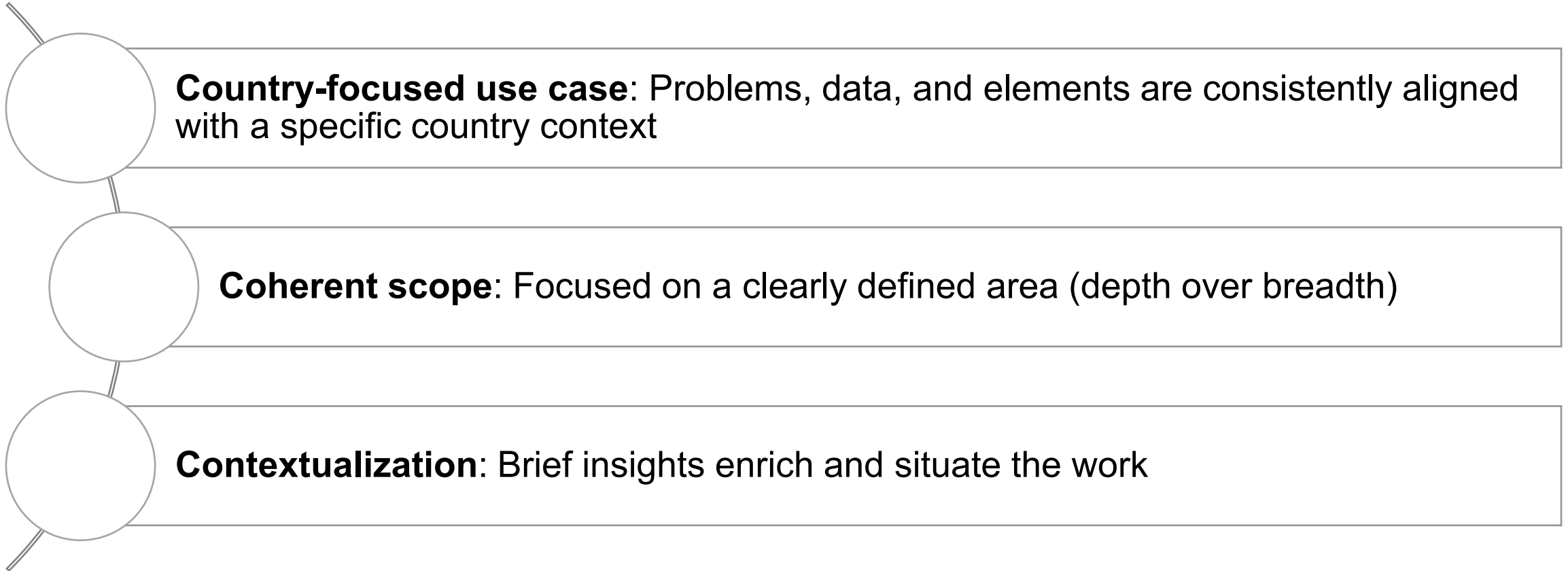
Analyze Performance

Item Properties Histograms



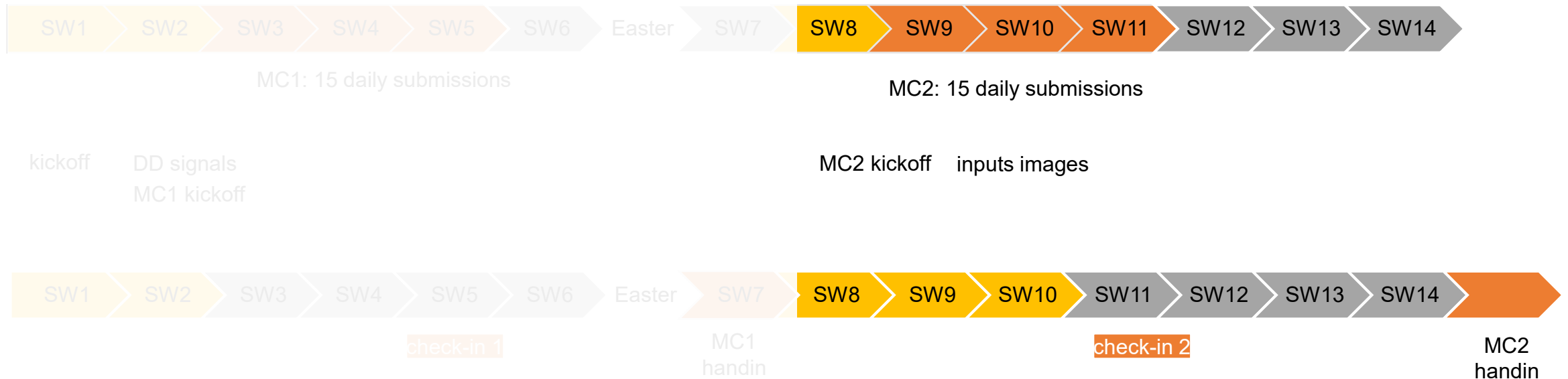
Formalities

Connecting the Dots with a Common Theme of a Country



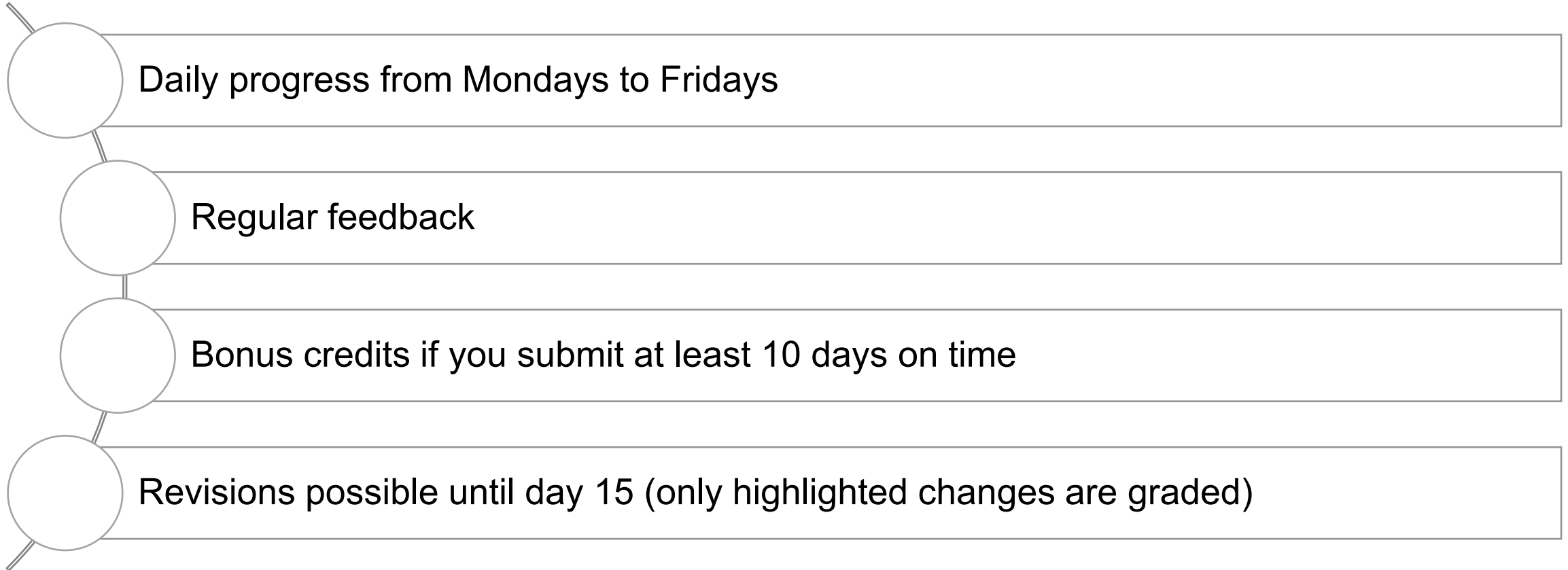
Mini-Challenge Versions

Version: 15-day-challenges



Version: individual MC rhythm

15-Day-Challenges Per Mini-Challenge



Check-Ins

- **Mandatory for individual MC rhythm**
- 15 min review meetings (see [Video: Warum Check-Ins \(German\)](#))
- One per MC and a third one if things don't go smooth
- During contact hours with subject matter expert → enrollment via [Calendly](#)
- Optional in 15-day-MC versions

Assignments - Submissions

- Submit HTMLs and link to sources in designated Teams assignment: E-gbsv-FS26_M365

E-gbsv-HS25_M365

Home page

Class Notebook

Classwork

Assignments

Grades

Reflect

Insights

Parents



Assignments

< Back

MC1 Day 13 - Sampling Theorem

Points

11 points possible

Due March 19, 2025 11:59 PM

Instructions

Instructions see attached word-document.

Submissions via Teams (click submit):

-HTML

-Link to code

Let's go and all the best for your submission!

Reference materials



gbsv_mc1_day13_fs25_2025-03-13.docx



Student work

None

Submit

Submission Requirements

- **Principle:** Individual, personalized work (no generic or shared submissions)
- **Artifacts**
 - Notebook (.ipynb) and HTML export (one notebook per subtask)
 - Accessible link to code and results (incl. signals/images, if applicable)
- **Format:** Language: German or English
- **Acceptance Criteria (mandatory):**
 - No ZIP files
 - No late submissions
 - Check-in completed (individual rhythm)
 - Submissions violating these are rejected

Grading

- According to 5 competences
- See grading rubric incl. gbsv interpretations and performance indicators
- MCs 1:1 ratio



How much time should I spend on daily submissions?

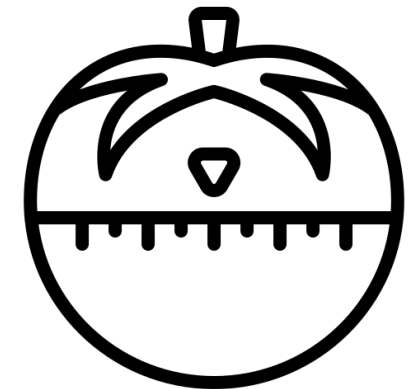
- Certain tasks are intentionally kept open-ended
- Set your own frame for limitations, or ask if the boundaries are unclear
- As a rule of thumb: one day's task should take around 2 hours (+/- 30 minutes)
- Extensive submissions/texts are neither required nor expected



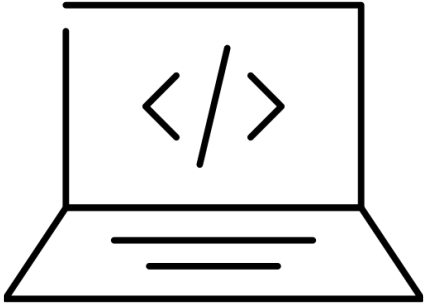
The Pomodoro Technique

- Work in focused intervals: 25 minutes (“1 Pomodoro”)
- Take a short break: 5 minutes after each Pomodoro
- Repeat: 4 pomodoros in a row
- Longer break: 15–30 minutes after 4 pomodoros
- Goal: Boost focus, reduce fatigue, and keep productivity high

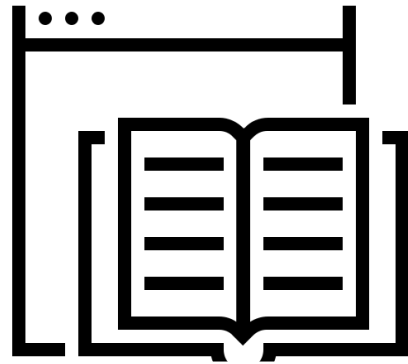
- MC:
 - Stop after 3-5 pomodoros - don't overwork in one go
 - Keep your submission time-boxed - focus on essentials
 - Revise later - e.g., on day 15 after a break



Suggested Resources: Your Swiss Pocket Knife for the MC



[Tutorial notebooks](#)



[Basics of Image
Processing Website](#)



[FAQ videos](#) (German)
[Image Processing
Foundations](#) (English)

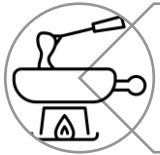


[Lesespuren](#): escape room
style short learning
quizzes (German)

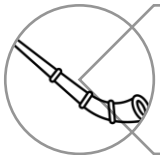
A Recipe for MC Quality, à la Swiss-made



Glide through your assignment like a skier: smooth, structured, and controlled



Match data, methods, and problems like fondue: a perfect fit



Let your creativity echo like an alphorn: focused and distinctive



Sharpen your analysis like a Swiss watch: precise and reliable



Keep it solid like Swiss cheese: clear, structured, and traceable

Acknowledgments

- Some image examples are from previous students taking the MC